

SMART INDIA HACKATHON 2026

Mausam home: actions with proof

Problem Statement Title-

Personalised homepage for the 'Mausam' mobile application

Problem Statement ID –

SIH26076

Theme-

Smart Automation

PS Category- Software/Hardware

Software

Team ID-

195697

Team Name (Registered on portal)

Zencoderss

Organisation

Ministry of Earth Sciences (IMD)

Every number shows its source, station, issue time and age. Every advisory can be re-derived by hand.



LIVE DEMO <https://mausam-sih26076-devanshdhangar70-7520s-projects.vercel.app/>

Guest-first PWA · works offline · English and Hindi



THE IDEA

One home screen that adapts to you — and shows its work

The Mausam home shows a number with no context: no station, no issue time, no age — so advice is trusted or ignored at random.

WHO IT AFFECTS

Commuters in fog, farmers at sowing, parents in a heat wave — anyone reading one number and acting on it.

WHAT'S BROKEN TODAY

A one-size homepage: readings shown without source, station, issue time or age.

WHY IT MATTERS

A wrong-day sowing or a late departure costs money and safety — and the app cannot be questioned.

Proposed Solution (Describe your Idea/Solution/Prototype)

A home screen that adapts to you: map on top, action cards below.

Detailed explanation of the proposed solution

Onboarding picks your persona; the home reorders — the farmer sees sowing first, the commuter sees fog first.

How it addresses the problem

Every card says what to do and shows the readings behind it: source, station, issue time and age.

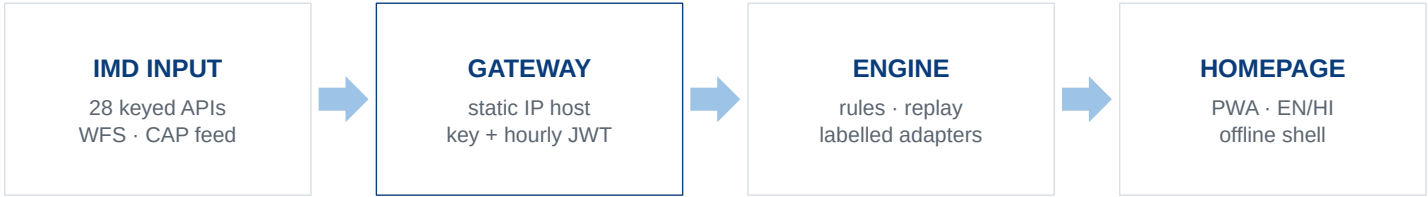
Innovation and uniqueness of the solution

Source on every number · replayable advice · honest gap cards · a grounded bot that never invents figures.

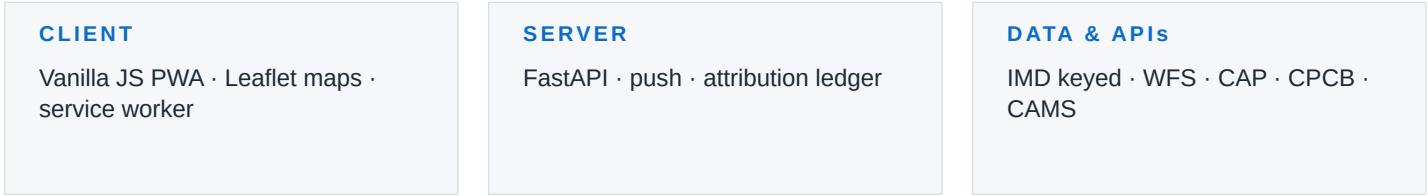
TECHNICAL APPROACH

IMD data in, verified decisions out

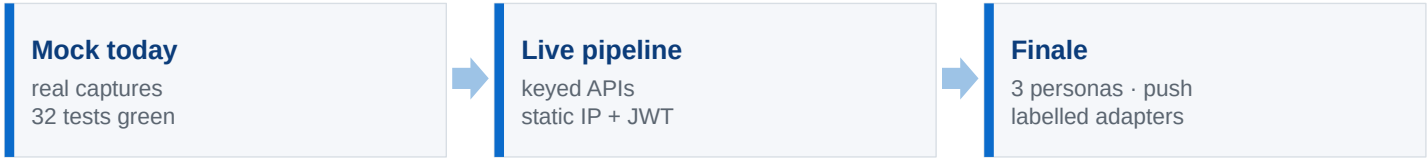
One deterministic pipeline: IMD keyed APIs → engine → decision stack, every value carrying its source, station, issue time and age.



Day-1 critical path

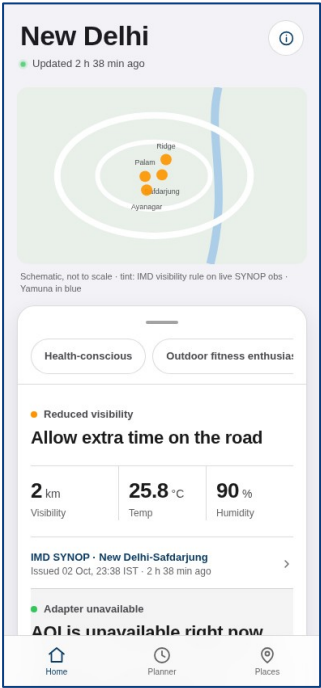


METHODOLOGY

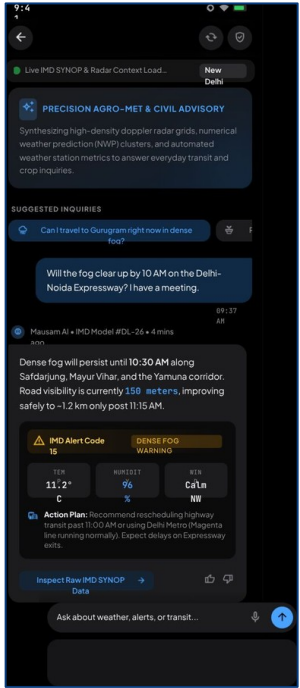


Technologies to be used (e.g. programming languages, frameworks, hardware)

Methodology and process for implementation (Flow Charts/Images/ working prototype)

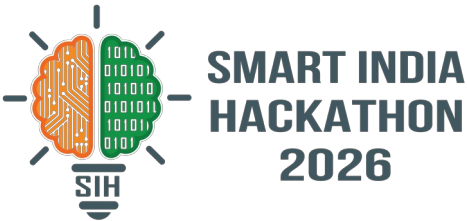


WORKING PROTOTYPE



ASSISTANT · DESIGN

The mock runs offline on captured IMD payloads; the keyed gateway is the Day-1 critical path.



FEASIBILITY AND VIABILITY

Every risk already has a working fallback

CHALLENGE

HOW WE HANDLE IT

Keyed API needs a static-IP host

Run on saved captures + public WFS meanwhile

CPCB feed may be unreachable

Labelled mirror fallback, else an honest gap card

Some IMD fields are undocumented

Documented fallbacks; the screen states its data grain

iOS limits push alerts

Demo on Android; the inbox keeps every alert

BUILD ORDER

Day 1

static-IP host + key registration

Week 1

captured-payload mock (done)

Week 3

live keyed pipeline

Finale

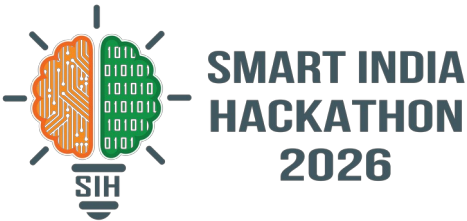
3 personas · push · labelled adapters

Viability: one container on a static-IP VM · one IMD API account (2 DEV + 2 PROD keys, attribution ledger on request) · no paid services · IMD data with attribution; CPCB/CAMS labelled adapters.

Analysis of the feasibility of the idea

Potential challenges and risks

Strategies for overcoming these challenges



IMPACT AND BENEFITS

Earlier action, traceable trust

Commuter

Fog window plus “leave 20 min early”, stated as an action, not a number.

WE MEASURE

Recall of the action after the alert

Farmer

Agromet sowing window and rainfall against normal, in the order he works.

WE MEASURE

Action recalled minutes later

Health

Heat card — and “IMD does not publish pollen” where the data is missing.

WE MEASURE

Readers treating the gap as a gap

BENEFITS

- **Earlier action** — each card states what to do and how fast.
- **Traceable trust** — every reading names its source, station and age.
- **Works offline** — the home shell keeps the last forecast.
- **Hindi + English** — warning text is human-reviewed, never raw machine translation.
- **Honest gaps** — missing data shows as a gap, never a fake zero.

Scalability — stateless engine, labelled adapters per source, personas as configuration: a new state needs data, not a rewrite.

Potential impact on the target audience	Benefits of the solution (social, economic, environmental, etc.)
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RESEARCH AND REFERENCES

Official sources, labelled adapters, no invented numbers

1. IMD API reference + portal user guide (key, hourly JWT, static IP)

api.imd.gov.in

2. CPCB real-time AQI, GODL

data.gov.in

3. Copernicus CAMS atmosphere data store

ads.atmosphere.copernicus.eu

4. MeitY UX4G design system + PIB Mausam note (2023)

ux4g.gov.in · pib.gov.in

5. Pai, N. et al. (2014), IMD Pune gridded rainfall 0.25°

[published dataset](#)

6. Mausam app: existing Damini lightning + Meghdoot crop modules

mausam.imd.gov.in

7. Working mock, fixtures, replay tests (32 pass)

[Mausam/docs/evidence](#) (illustrative)

8. Live demo – deployed PWA, guest-first, works offline

mausam-sih26076-devanshdhangar70-7520s-projects.vercel.app/

How we differ — Damini and Meghdoot show values; we show values with source, station, issue time and age, plus replayable advice and honest gap cards.

Details / Links of the reference and research work

SOURCE MAP

IMD official · keyed API

CPCB official · labelled adapter

Copernicus CAMS official · labelled

MeitY UX4G design standard

Our mock 32 tests · replay



Scan to try the live demo

<https://mausam-sih26076-devanshdhangar70-7520s-projects.vercel.app/>

Guest-first PWA · offline · EN/HI

Official, verifiable sources; local mock paths are labelled (illustrative).
No fabricated statistics.